

# Nguyen Nguyen

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Software engineer focused on backend systems and data-intensive applications, with research training in optimization and machine learning. Experienced in designing versioned data models, building Dockerized services, and deploying ML pipelines used in production clinical and research environments.

## PROJECTS

### Mokuro Library

11/2025 - Present

- Built a self-hosted, multi-user Japanese OCR reader and editor for NAS/home servers, with authentication, binary file streaming, and reproducible Docker Compose deployment.
- Designed a database-backed OCR patch tree with copy-on-write branching, enabling persistent document versioning, snapshot caching, and efficient state reconstruction via patch replay.
- Implemented collaborative editing workflows on shared content: users read an admin-maintained baseline by default, with first edit forking into a private branch supporting undo/redo, reset, admin merge, and rebase with conflict resolution.
- Built a client-side editing engine for OCR text and bounding boxes, implementing zoom-invariant geometry transforms, keyboard-driven text mutation, and Japanese vertical/horizontal layout semantics in the browser.
- Designed a hybrid shared-library system combining public admin-owned content with private user uploads, per-user reading state, and a submission/review workflow for promoting contributions.

### Quantum Virtual Machine (QViM)

03/2021 - 05/2021

- Created a Domain-Specific Language in Julia using meta-programming to define quantum circuits with readable syntax.
- Implemented a circuit simulator that optimizes gate operations via linear algebra basis changes to reduce computational overhead.

## WORK EXPERIENCE

### Paschalidis NOC Lab

Graduate research assistant | Full-time

05/2022 - Present  
Boston, MA

- Formulated a novel spectral algorithm for discovering latent policies from demonstrations, guaranteeing global convergence in a single data pass and overcoming EM method pitfalls.
- Designed and implemented a flexible framework for discovering latent processes, with applications in bioinformatics (mutation analysis) and remote sensing (hyperspectral unmixing).
- Built and deployed an ML pipeline in 6 months, trained on a 100,000-patient dataset and is now running weekly inference for BMC's clinical trial for hypertension prescription.
- Built a data pipeline for a complex 10-year dataset of 30,000 appointments for 6,000 patients to power ML prediction of missed CT screenings, enabling targeted patient support.

## SKILLS

**Programming languages:** Python, TypeScript, Bash, Rust, Julia, Lua, SQL, C++

**Technical skills:** large-scale data processing, exploratory data analysis, data integration, statistics, probability, optimization, machine learning, dynamic programming, predictive modeling, data structures

**Framework and libraries:** PyTorch, Hugging Face, pandas, Weights & Biases, cvxpy, pydantic, scikit-learn, Jupyter, SvelteKit, Prisma, Fastify

**Tooling:** Cargo, Conda, Linux/Unix, Nix, Git, Latex, Docker

**Languages:** English (fluent), Vietnamese (native), Japanese (conversational)

## EDUCATION

**PhD. in Systems Engineering** | Boston University

Boston, MA | 08/2021 - 06/2026

**B.S. in Computer Science** | Syracuse University

Syracuse, NY | 08/2017 - 05/2021

**B.A. in Physics** | Syracuse University

Syracuse, NY | 08/2017 - 05/2021